

Cambridge established women's colleges around 1870, but although female students could attend lectures and take exams, they were not awarded degrees until 1920 at Oxford—and 1948 at Cambridge.

Scientific revolution

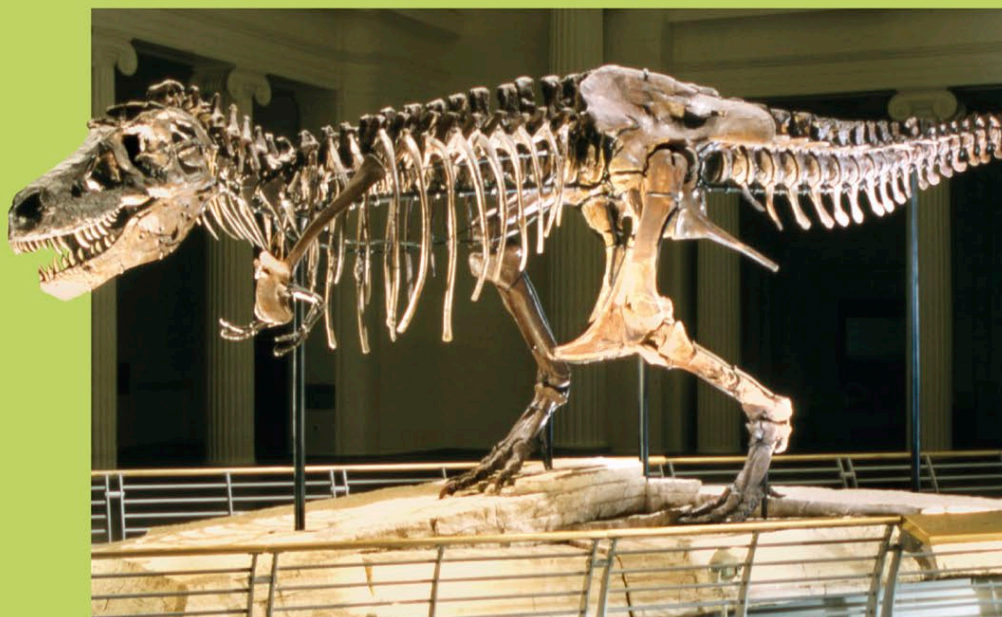
In the 20th century, the number of women in science exploded. From Gerty Cori, whose work in biochemistry led to discoveries about human metabolism, to Chien-Shiung Wu and Lise Meitner, who made breakthroughs in nuclear fission in the 1930s and 1940s, women were behind some of the century's key innovations. Despite these advances, women often saw the credit for their work go to male scientists. Meitner, along with chemist Isabella Karle, and astrophysicist Jocelyn Bell Burnell were overlooked for Nobel Prizes, while their male team members were honored.

Nowadays, the story is very different. Women work as engineers, researchers, nuclear physicists, and more. Today's female science and mathematics students will continue to change the world, but they owe a huge debt to the brave women who led the way.



▼ Scary Sue

The world's largest complete *Tyrannosaurus rex* skeleton is named Sue after the paleontologist Sue Hendrickson, who discovered the fossil in South Dakota in 1990.



PROFILES

STEM SCHOLARS

Hypatia of Alexandria (c. 355–415) studied under her father, a noted mathematician and astronomer. Although Alexandria was under Roman rule, Hypatia worked to preserve the works of Greek mathematicians and astronomers. She produced works on astronomy and geometry, but they have been lost. During her life she taught widely and gave popular lectures on many topics.

Wang Zhenyi (1768–1797) was taught astronomy, medicine, and mathematics by her older relatives, as well as poetry. An avid reader, Wang wrote a simplified version of the famous mathematics book *Principles of Calculation*, intending it for female and male students. Her astronomical work included calculating the movement of stars and planets, and explaining the causes of a lunar eclipse.

Caroline Herschel (1750–1848) trained as a singer but she eventually gave up her musical career to help her brother William with his astronomical research. She ground mirrors for his telescopes and performed the calculations needed for observing the skies. Once she got her own telescope she scanned the skies, discovering three nebulae and eight comets.

Sofia Kovalevskaya (1850–1891) was refused a place at a Berlin university because she was a woman, so she studied privately instead. Her work on differential equations won her recognition in mathematics, and she later taught at the University of Stockholm. She was the first woman in modern Europe to earn a doctorate in mathematics and the first to be given a full professorship.

Rosalind Franklin (1920–1958) was a pioneer in the use of X-ray diffraction technology. She used it to study DNA at a time when the chemical makeup and structure of DNA was still unknown. She showed that it had a helical shape, producing X-ray patterns of the complicated molecule that allowed Nobel Prize winners Watson and Crick to confirm its structure.

Katherine Johnson (b. 1918) began her career as a teacher, at a time when few professions were open to African American women. She got a job analyzing data and performing computations for the space program. Her work helped calculate the flight paths of spacecraft, including the Apollo 11 mission to the moon. Her story featured in the acclaimed 2016 movie *Hidden Figures*.